

Urinary Problems After Pelvic Radiation

Patient Information · Late effects on the bladder and urethra, and how they are treated

WARWIKI

Radiation for prostate, cervical, uterine, rectal or bladder cancer can affect the bladder and urethra. Some problems start during treatment and settle within weeks. Others appear **months or many years later**, as radiation slowly harms small blood vessels and leaves stiff scar. Most can be treated, and **reconstructive surgery** can help when simpler care does not.

What Can Happen

- **Radiation cystitis:** the bladder lining becomes fragile. It can cause urgency, going often, pain and **blood in the urine**, from 6 months to 20 years after treatment.
- **A small, stiff bladder** that holds less, so you go very often or leak.
- **A narrowing** of the urethra or bladder outlet, with a weak stream and straining.
- **Leaking** from a weak control muscle, a small bladder, or both.
- **A fistula**, an abnormal opening from the bladder or urethra into the rectum, vagina, skin or pubic bone. It is less common.

Benefits and Risks in Numbers

about
2 in 100

got a narrowed urethra within about 4 years of prostate radiation (5 in 100 after seeds plus beam)

about
1 in 2

with radiation bleeding had it stop fully after hyperbaric oxygen

about
8 in 10

urethra repairs after radiation worked. About 1 in 5 had new leaking

Numbers are averages from published studies. Your own risk may differ; ask your team.

Hyperbaric Oxygen

You breathe pure oxygen in a pressurized chamber for about 1½ hours a day, 5 days a week, for 30 to 40 sessions. It helps damaged tissue heal. In a trial, it eased bladder symptoms more than usual care. Starting it early seems to help more. Ear pressure and short-term blurry vision are common. It does not stop heavy bleeding in an emergency.

Learn the Terms

Radiation cystitis

Bladder irritation and bleeding caused by past radiation.

Hyperbaric oxygen

Breathing pure oxygen under pressure in a special chamber.

Stricture, stenosis

A narrowing of the urethra or bladder outlet caused by scar.

Fistula

An abnormal opening between two organs that should not be joined.

Cystoscopy

A look inside the bladder with a thin camera.

Bladder washout

Rinsing clots out of the bladder through a catheter.

Embolization

Blocking the small artery that is bleeding, through a thin tube in a blood vessel.

Urinary diversion

Rerouting urine away from the bladder, often to a bag on the belly.

Is blood in my urine always from radiation? No. Even years after radiation, blood in the urine needs a check of the bladder and kidneys. Radiation raises the chance of a new bladder cancer later, so the cause must be found before it is blamed on radiation.

During treatment

Burning and going often are common. They usually settle within a few months.

Months to years later

Bleeding, urgency, leaking or a slow stream can start.

When it starts

Tell your team. Tests find the cause and rule out other problems.

Treatment

Step by step, from simple care to surgery when needed.

How It Is Treated

Bleeding

A catheter and washout to clear clots, a camera to seal bleeding spots, and medicine put into the bladder. Hyperbaric oxygen for bleeding that keeps coming back. Embolization or surgery for severe bleeding.

Narrowing

Stretching or cutting through a scope, or reconstruction (urethroplasty) at a specialist center. A sphincter implant may be needed later for leaking.

Fistula

Repair with healthy tissue moved in from nearby, or rerouting urine (and sometimes stool) away from the damaged area.

A badly damaged bladder

When the bladder is very small, painful or keeps bleeding, urinary diversion, with or without removing the bladder, can relieve pain and bleeding.

Call your care team if you have:

- A temperature of **100.4 °F (38 °C)** or higher, or chills
- New blood in the urine, or a stream that is slowing
- Air or stool in the urine, urine from the rectum or vagina, or new pain in the pubic bone, groin or hip

Go to an emergency department if:

- You cannot urinate for several hours and your bladder feels full or painful
- Your urine looks like red wine or has clots, or a catheter stops draining
- You feel faint or dizzy with bleeding, or cannot reach your team and are worried

Three Things to Remember

1. Urinary problems from radiation can start many years after treatment.
2. Blood in the urine always needs a check, even after radiation.
3. If simple treatment does not help, ask to see a reconstructive specialist.

What to Bring to Your Visit

- Your radiation details: where, when and what type (beam, seeds or both)
- A list of past procedures, blood thinners, and how many pads you use
- Your latest cancer results